Safety With Electricity

Below are two press articles which refer to electrical safety. Read the articles and answer the questions which follow. You may need to refer to other sources for some of the answers.

Schoolgirl Dies

An 11-year-old Sydney schoolgirl has died from injuries received in a pool accident which also killed her best friend. A spokesman for Westmead Hospital said she died early yesterday morning. She never recovered from receiving electric shocks in a swimming pool on 19 December. Her friend, also 11, died later that same day. A live wire came into contact with the water in which the two girls were swimming after the pool filter's electrical cable was accidentally cut.

The Age, 27 December 1988

Electrical Shocks More Likely in Summer

By DAVID THOMSON

People are twice as likely to suffer electrical shocks in summer as in the rest of the year, the SEC has warned.

A survey of the number of shocks reported in the 10 years to 1986 found that 65 per cent of them occurred between October and March, the SEC's acting Victorian Chief Electrical Inspector, Mr Ian Graham, said.

He said the risk increased in summer because portable appliances were used outdoors and in tents and caravans, people were barefooted or near water, especially pools, and people tried to do their own electrical repairs and installations.

Mr Graham said 500 electrical shocks had been reported to the SEC last year. There had been another 100 cases in council supplied areas. About one-tenth of all shocks were the result of dampness.

He said there had been 17 fatalities from electrical appliances in 1987. There have been two deaths from electrical contact with swimming pools since wiring regulations were revised in 1976 to isolate electrical equipment, including earthing and lighting, from pools.

Mr Graham said checking existing electrical fittings was much more difficult than issuing guidelines for new works.

"We cannot check every single pool in Melbourne; we do not have the right to interfere."

Mr Graham said pool owners should check permanently wired electrical fittings such as pumps and underwater lights. Variations in water levels and heat caused power lines to sag, which was dangerous where they crossed water.

He said there had been a number of fatalities caused by hair dryers falling into baths. Mr Graham said people should remember that it was illegal and unsafe for unlicensed tradesmen to work on fixed wiring of houses, and that caution was needed with extension cords to tents and caravans.

Questions

1 How might the accident in which the two school girls were killed, have been avoided ?

2 List the reasons why it is more likely that more electric shocks are experienced in summer.

3 How could a hair drier falling into a bath cause a fatality ?

4 Why would dampness contribute to a large number of electrical shocks ?